

END SEMESTER ASSESSMENT (ESA) - Jan - May 2024**UE21CS351B - Cloud Computing****Total Marks : 100.0**

1.a. How is scalability different from elasticity in the cloud? What type of scalability is present in cloud? Discuss the type that you chose.
"Private cloud do not need to address the challenge of elasticity" - state whether this statement is true/false and justify the statement. (10.0 Marks)

1.b. Discuss the relative advantages and disadvantages of private and public clouds. Mention the tools/technologies that illustrate private and public cloud features or examples (10.0 Marks)

1.c. Discuss the three cloud service models with suitable examples. (5.0 Marks)

2.a. Discuss different types of hypervisors. Give an example of each.
How paravirtualization and full virtualization techniques are used to virtualize x86. (10.0 Marks)

2.b. Consider an architecture which supports the following instructions

Instruction	Behaviour User Mode	Behaviour PrivilegeMode
move <u>rA</u> , <u>rB</u>	moves content of register A to register B for all <u>General purpose</u> registers	moves content of register A to B for all <u>general purpose</u> registers
<u>loadpgTable</u> <u>rA</u>	Generates a trap	moves the contents of <u>rA</u> into the page table register
int x	generate a software or hardware int after moving to privilege mode	Not permitted
<u>clearInt</u>	will fail but generate no error	clears flag so no interrupts are possible

(i) mark each of the instructions as to whether they are sensitive and if so whether they are behaviour or control sensitive(6). (ii) based on this data will you be able to design a trap-and-emulate hypervisor for this architecture. Justify your solution (10.0 Marks)

2.c. How do you differentiate between container and Virtual Machines. Mention example tools/technologies for containers and virtual machines. (5.0 Marks)

3.a. What is a microservice application and how is it different from a regular monolithic application? How is REST related to the microservice programming model? Discuss REST in brief. (10.0 Marks)

3.b. Bring out the comparison between hot migration and cold migration? Explain the different copy techniques of hot migration. (10.0 Marks)

3.c. You are asked to design a multitenant database for two hospitals – *BestCare* and *PatientFriendly* to store information about patients. *BestCare* wants to store PatientID, names and previous history while *PatientFriendly* wants to store PatientID, names and NextAppointment date. Design a multitenant database using the preallocated column method for the same. (5.0 Marks)

4.a. Discuss the rebalancing of partitions in cloud storage. Bring out any of the hash based partitioning methods. Discuss the leaderless replication method to keep copies of data in cloud storage. (10.0 Marks)

4.b. Which are the three major leader election algorithms in cloud systems. Discuss the relative merits and demerits of Ring algorithm along with details on message latencies. (10.0 Marks)

4.c. What is a reverse proxy? What additional features can it provide. (5.0 Marks)